

Page Layout

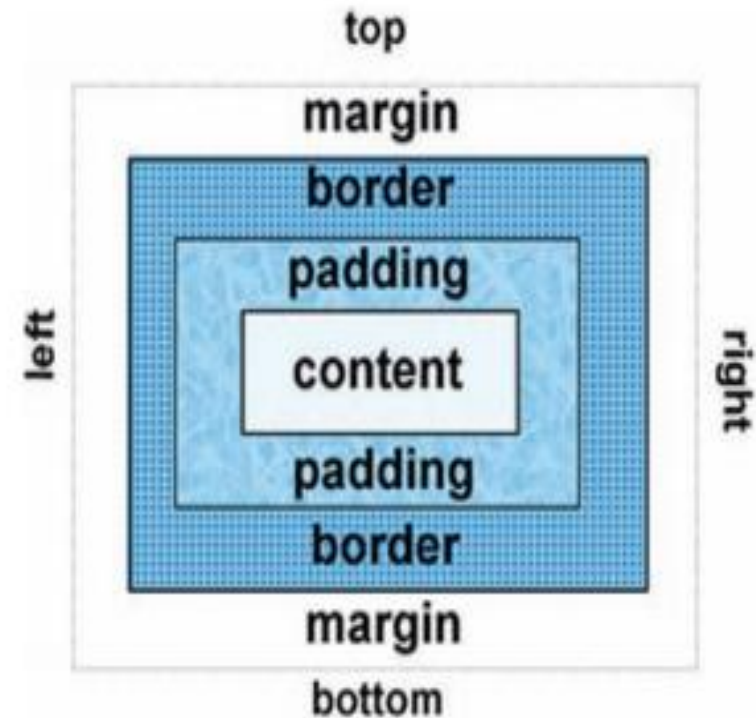
Chapter-7

Contents

- The Box Model
- Normal Flow
- CSS Float
- CSS Two-Column Layout
- Hyperlinks in an Unordered List
- CSS pseudo-classes
- Positioning with CSS

The Box Model

- **Content**-Text and web page elements in the container
- **Padding**-Area between the content and the border
- **Border**- Between the padding and the margin
- **Margin**- Determines the empty space between the element and adjacent elements



Margin Property

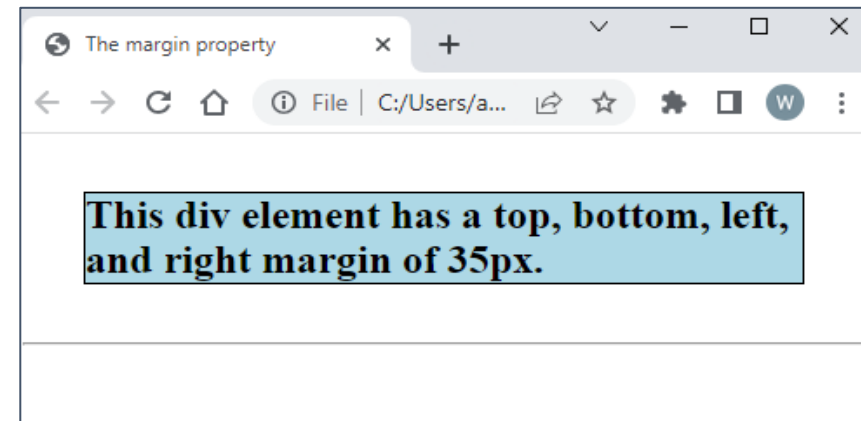
Property	Description	Example
margin	configure the margin surrounding an element A numeric value (px or em), percentage, or auto;	margin: 10px; margin: 10px 30px; margin: 10px 30px 20px; margin: 10px 30px 20px 5px;
margin-top	top margin;	margin-top:20px;
margin-right	Right margin;	margin-right:20px;
margin-bottom	Bottom margin;	margin-bottom:20px;
margin-left	Left margin;	margin-left:20px;

Sample Code and Example

```
<!DOCTYPE html>
<html>
<head>
<title>The margin property</title>
<style>
h2 { margin: 35px;
    border: 1px solid black;
    background-color: lightblue;
}
```

```
</style>
</head>
<body>
<h2>This div element has a top, bottom, left,
and right margin of 35px.</h2>

</hr>
</body>
</html>
```



Padding Property

Property	Description
padding	A shorthand property for setting all the padding properties in one declaration
padding-bottom	Sets the bottom padding of an element
padding-left	Sets the left padding of an element
padding-right	Sets the right padding of an element
padding-top	Sets the top padding of an element

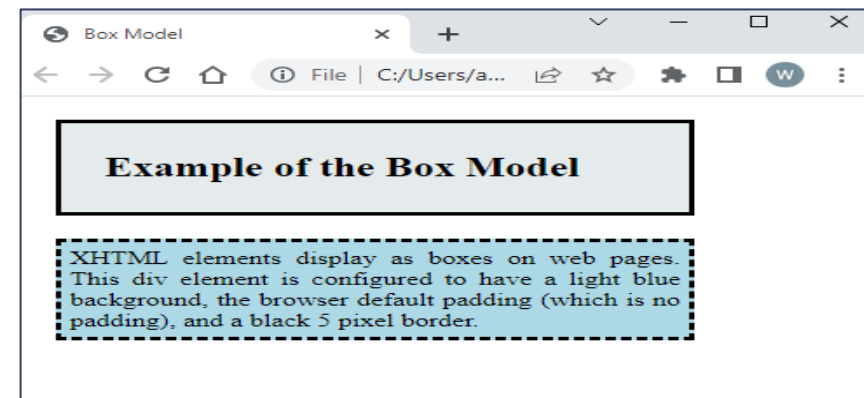


Sample Code and Example

```
h2.ex{  
margin:20px; width:310px;  
padding: 25px;  
border-style: solid;  
background-color: rgb(229, 235, 236); }  
  
div { margin:20px; width: 350px;  
padding: 5px; border-style: dotted;  
background-color: lightblue;  
text-align: justify; }
```

<h2 class=" ex">Example of the Box Model</h2>
<div>XHTML elements display as boxes on web pages. This div element is configured to have a light blue background, the browser default padding (which is no padding), and a black 5 pixel border.

</div>



Normal flow

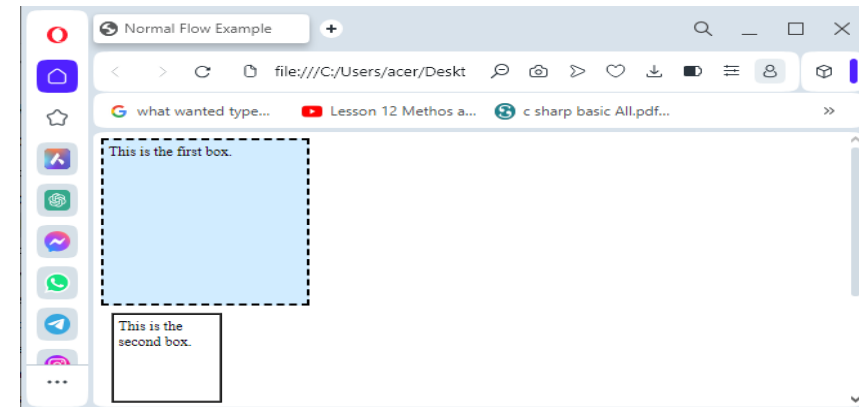
- The browser used normal flow (**the default**) and displayed the elements in the order that they appeared in the source code.

```
.div1 { width: 200px; height: 200px;  
background-color: #D1ECFF;  
border: 3px dashed #000000; padding: 5px;  
}
```

```
.div2 { width: 100px; height: 100px;  
background-color: #ffffff;  
border: 3px ridge #000000; margin: 10px;  
padding: 5px; }
```

```
<div class="div1"> This is the first box.  
</div>
```

```
<div class="div2">This is the second box.  
</div>
```

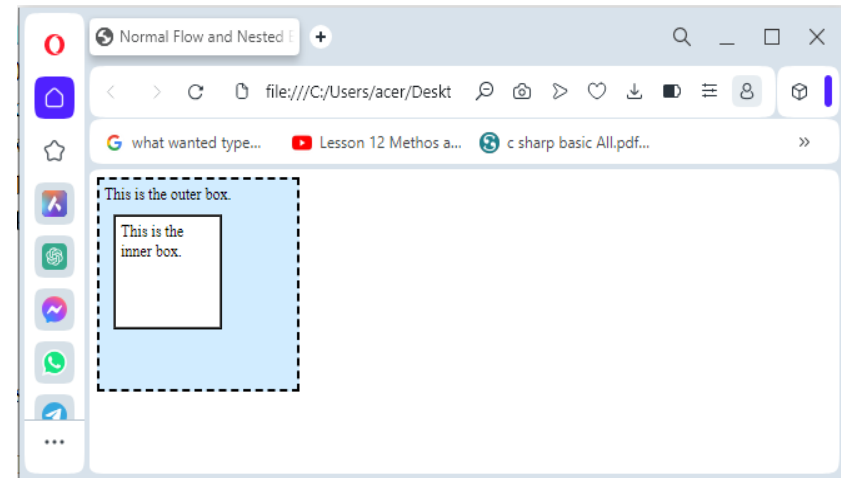


Normal flow & Nested Elements

```
.div1 { width: 200px; height: 200px;  
background-color: #D1ECFF;  
border: 3px dashed #000000; padding: 5px;  
}
```

```
.div2 { width: 100px; height: 100px;  
background-color: #ffffff;  
border: 3px ridge #000000; margin: 10px;  
padding: 5px; }
```

```
<div class="div1"> This is the first box.  
<div class="div2">This is the second box.  
</div>  
</div>
```



CSS Float

- Elements that seem to float on the **right** or **left side** of either the browser window or another element are often configured using the **float property**.
 - float-right
 - to float the element on the right side of the container.
 - float-left
 - to float the element on the left side of the container.

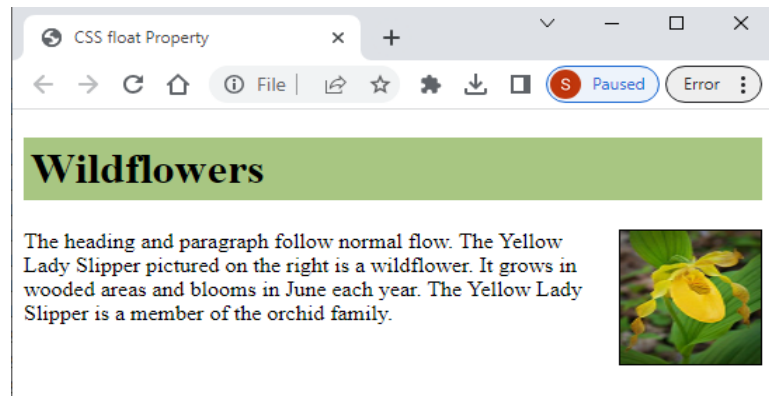
Overflow Property

- The **overflow property** is often used to **clear a float**.

value	Description
visible	Default value ; the content is displayed , and if it's too large, the content will overflow and extend outside the area allocated to it
hidden	The content is clipped to fit the area allocated to the element in the browser viewpoint
auto	The content fits the area allocated to it and, if needed , scrollbars are displayed to allow access to the remaining content
scroll	The content is rendered in the area allocated to it and scrollbars are displayed

Sample Code and Example

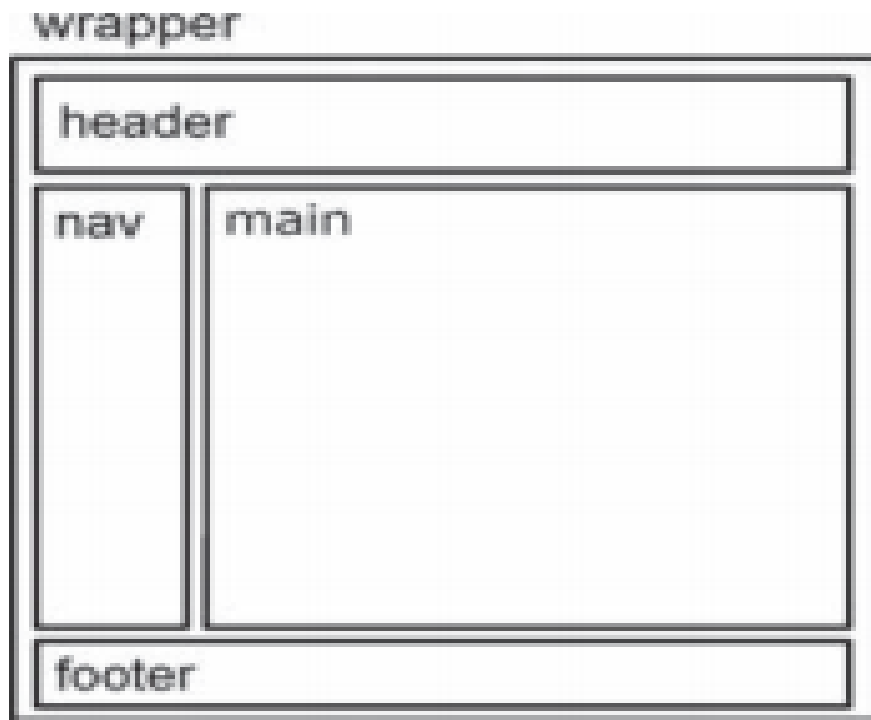
```
h1 { background-color: #A8C682;
padding: 5px; color: #000000; }
p { font-family: Arial, sans-serif;
overflow: visible; }
img { float: right; margin: 0 0 5px;
border: 1px solid #000000; }
```



```
<h1>Wildflowers</h1>

<p>The heading and paragraph follow normal flow.
The Yellow Lady
Slipper pictured on the right is a wildflower. It grows
in wooded
areas and blooms in June each year. The Yellow Lady
Slipper is a
member of the orchid family.</p>
```

CSS Two-Column Layout



Two-column layout

```
<body>
```

```
<div id="wrapper">
```

```
    <header> </header>
```

```
    <nav> </nav>
```

```
    <main> </main>
```

```
    <footer> </footer>
```

```
</div>
```

```
</body>
```

Hyperlinks in an Unordered List

Vertical navigation menu

```
<nav><ul>
```

```
<li><a href="index.html">Home</a></li>
```

```
<li><a href="menu.html">Menu</a></li>
```

```
<li><a href="directions.html">Directions</a></li>
```

```
<li><a href="contact.html">Contact</a></li>
```

```
</ul></nav>
```

Configure with css

```
nav ul { list-style-type: none; }
```

- [Home](#)
- [Menu](#)
- [Directions](#)
- [Contact](#)

Remove the Underline with CSS

```
nav a { text-decoration: none; }
```

[Home](#)
[Menu](#)
[Directions](#)
[Contact](#)

Home
Menu
Directions
Contact

Configure List Markers with CSS

CSS properties for ordered and unordered list

Property	Description	Value	List Marker Display
list-style-type	Configures the style of the list marker	none	No list markers display
		disc	Circle
		circle	Open circle
		square	Square
		decimal	Decimal numbers
		upper-alpha	Uppercase letters
		lower-alpha	Lowercase letters
		lower-roman	Lowercase Roman numerals
list-style-image	Image replacement for the list marker	The url keyword with parentheses surrounding the file name or path for the image	Image displays in front of each list item
list-style-position	Configures placement of the list marker	inside	Markers are indented; text wraps under the markers
		outside (default)	Markers have default placement

Display Property

- The CSS **display property** configures how the browser renders, or displays, an element on a web page.

value	Description
none	The element will not display
inline	The element will display as an inline element in the same line
Inline-block	The element will display as an inline element adjacent to another inline display elements but also can be configured with properties of block display elements including width and height .
block	The element will display ad an a block element

Sample Code and Example

Horizontal Navigation Menu

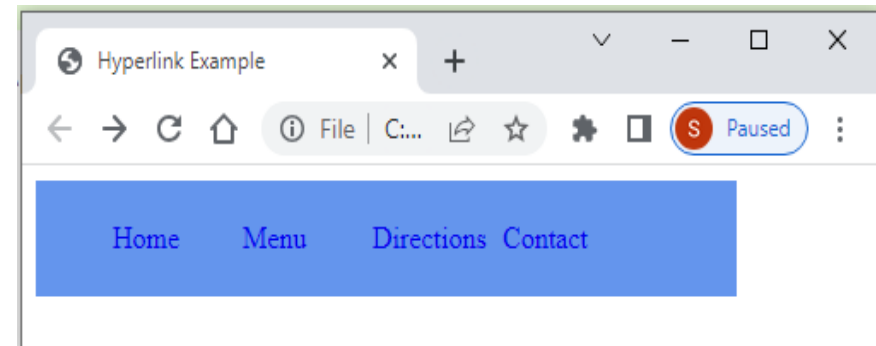
```
<nav>
<ul> <li><a href="index.html">Home</a></li>
<li><a href="menu.html">Menu</a></li>
<li><a href="directions.html">Directions</a></li>
<li><a href="contact.html">Contact</a></li>
</ul>
</nav>
```

Configure with CSS

```
nav ul { list-style-type: none; }
```

```
nav li { display: inline; }
```

```
nav a { text-decoration: none; padding-right: 10px; }
```



CSS pseudo-classes

- A pseudo-class is used to define a **special state** of an element.

syntax

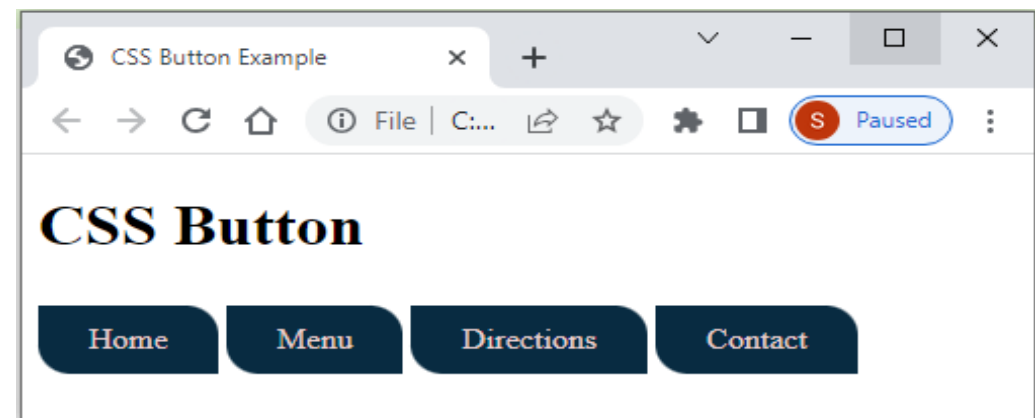
`selector : pseudo-class { property: value; }`

Pseudo-Class	When Applied
:link	Default state for a hyperlink that has not been clicked (visited)
:visited	Default state for a visited hyperlink
:focus	Triggered when the hyperlink has keyboard focus
:hover	Triggered when the mouse pointer moves over the hyperlink
:active	Triggered when the hyperlink is actually clicked

CSS Button Sample Code and Example

```
a {  
  background-color: #082b41;  
  border-top-right-radius: 1em;  
  border-bottom-left-radius: 1em ;  
  color: rgb(235, 208, 208);  
  padding: 10px 25px;  
  text-align: center; font-size: 16px;  
  text-decoration: none;  
  display: inline-block; margin: 4px 0px;  
  cursor: pointer; }
```

```
<h1>CSS Button</h1>  
<a href="index.html">Home</a>  
<a href="menu.html">Menu</a>  
<a href="directions.html">Directions</a>  
<a href="contact.html">Contact</a>
```



Positioning with CSS

- The position property specifies the type of positioning method used for an element.

Value	Description
static	Default value ; the element is rendered in normal flow
fixed	Configures the location of an element within the browser viewport; the element does not move when the page is scrolled
relative	Configures the location of an element relative to where it would otherwise render in normal flow
absolute	Precisely configures the location of an element outside of normal flow

- **Note :** The top, right, bottom, and left properties are used to position the element.

Sample Code and Example

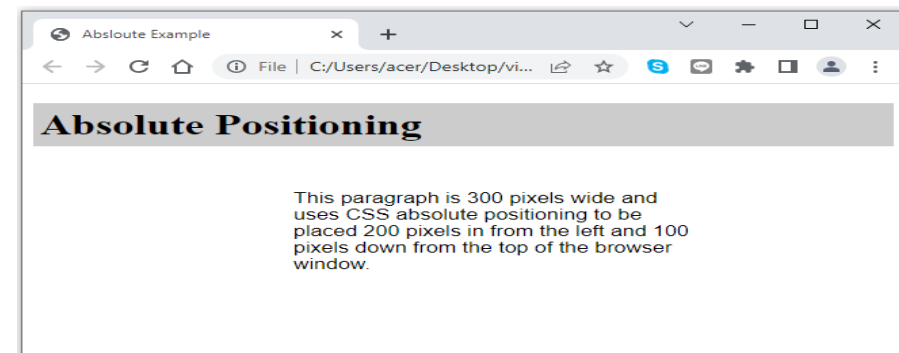
```
p { position: absolute;  
  
left: 200px;  
  
top: 100px;  
  
font-family: Arial, sans-serif;  
  
width: 300px; }
```

<h1>Absolute Positioning</h1>

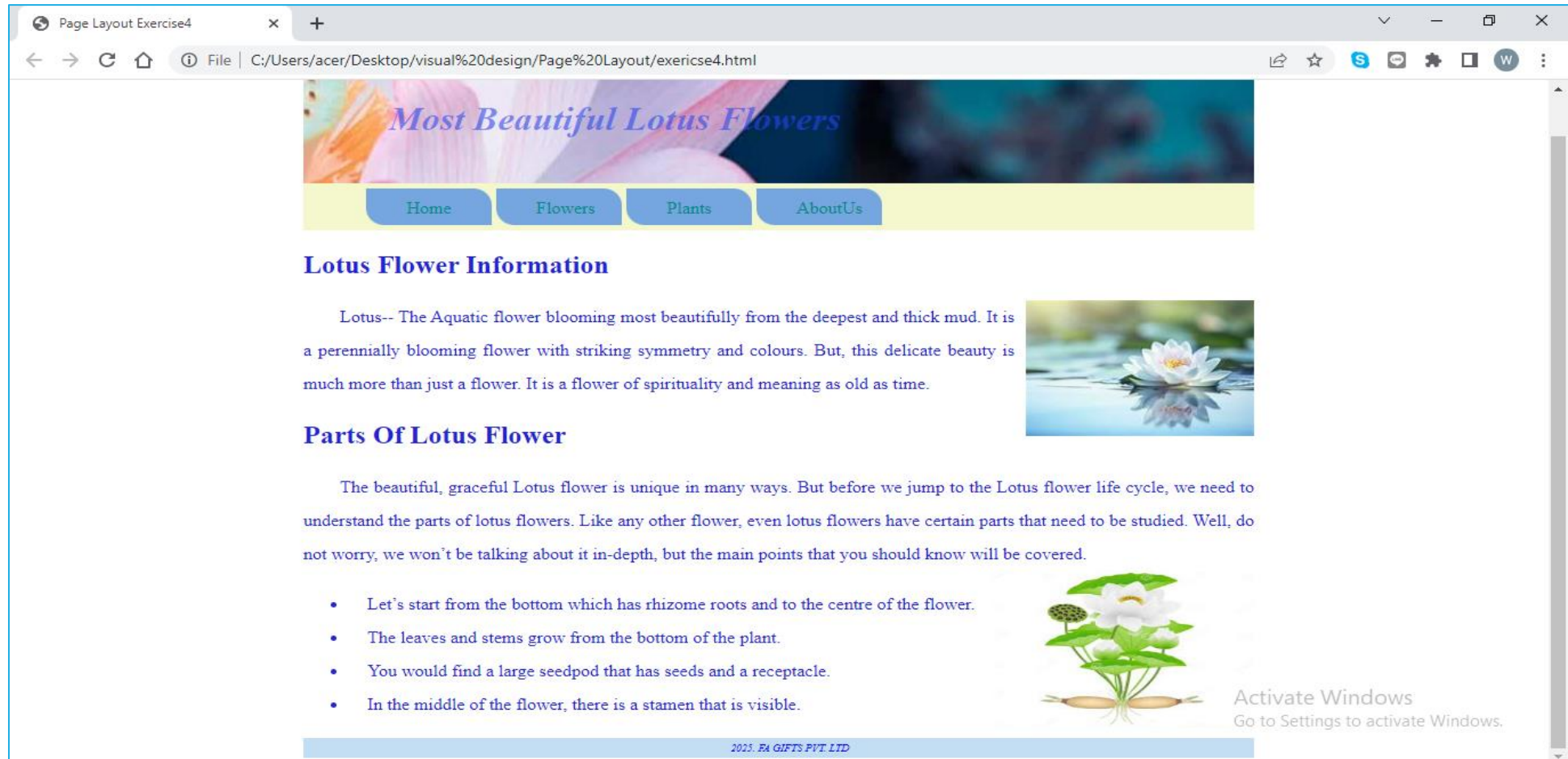
<p>This paragraph is 300 pixels wide and uses CSS absolute

positioning to be placed 200 pixels in from the left and 100 pixels

down from the top of the browser window.</p>



Exercise 1



Next Lecture

Tables